CSMFAB000000000113-Vol-17

Orbital Velocity Gap Analysis: What Second Stage is Needed?

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1. Delta-V Budget for Each Scale

1.1 The Tyranny of the Rocket Equation

The Tsiolkovsky rocket equation:

`delta_v = Isp * g0 * ln(m0 / m_dry)`

Where Isp = specific impulse (s), g0=9.81 m/s2, m0=initial wet mass, m_dry=dry mass.

For a solid rocket motor (SRM) with Isp=260s (typical composite propellant): Mass ratio required = e^(delta_v / (Isp * g0)) = e^(7177 / (260 * 9.81)) = e^2.81 = 16.6

The second stage must be 94% propellant by mass. This is aggressive but achievable.

1.2 Second Stage Mass Requirements by Scale

Scale	v_exit (m/s)	dv_needed (m/s)	Mass ratio	Stage mass (kg)	Propellant mass (kg)
Nano	340	~7500	18.2	0.25	0.237
Micro	690	~7200	16.4	3.5	3.29
Small	1380	~6600	13.3	35	32.4
Medium	1040	~6900	14.9	350	327
Large	680	~7200	16.4	4000	3756

Scale 3 (Small) is the most efficient: 35 kg total second stage for 5-10 kg payload. This is in the range of commercially available hobby SRMs (e.g., AeroTech K-class: up to 2500 $N \cdot s$, $80-$400).

1.3 Effective Payload Mass Fractions

If total launch vehicle (spear + payload + second stage) is 50 kg: - Scale 3 (Small): Payload = 50 - 10 (spear) - 35 (stage) = **5 kg payload** (10%) - Scale 4 (Medium): Payload = 500 - 100 - 350 = **50 kg payload** (10%) - Scale 5 (Large): Payload = 5000 - 1000 - 4000 = **0 kg** (negative! impossible at this scale)

Large scale cannot reach orbit without multi-staging or increased exit velocity.

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2. Two-Stage Configuration

2.1 The Ideal Staging Strategy

The AMSR-UBSCLS constitutes Stage 0 (pre-boost). The second stage fires at apogee:

Optimal firing time: At apogee, where v_horizontal is maximum fraction of total velocity.

For a 42° launch at v_exit=1462 m/s: At apogee (102 km): - v_horizontal = v_exit * cos(42°) * (1 - drag_loss_h) = 1086 * 0.82 = 891 m/s - v_vertical = 0 (by definition at apogee)

Required circularization burn at 102 km altitude: v_orbital(102km) = sqrt(mu / (R_earth + 102km)) = sqrt(3.986e14 / 6.478e6) = 7843 m/s

Delta-v for circularization: 7843 - 891 = **6952 m/s**

2.2 Miniaturized SRM Design (GIC-Immune)

The second stage must be GIC-immune per Aegis protocol: - Case: BFRP/Elium® filament-wound pressure vessel (same as CSMFAB000000000009) - Propellant: APCP (ammonium perchlorate composite) — no metallic components - Igniter: KNbO3-BaTiO3 piezoelectric shock igniter (from CSMFAB000000000021 EMF detector tech) - Nozzle: ZrB2-SiC ceramic (3245°C melt vs. ~3500°C combustion gas temp — marginal; see Vol 20)

SRM for Scale 3 (35 kg total mass, 32.4 kg propellant, 2.6 kg case/nozzle): - Burn time: ~30 s at average thrust - Average thrust: F_avg = 32.4 * 240 / 30 = 260 N (Isp=260s gives exhaust velocity 2550 m/s) Wait: thrust = m_dot * v_exhaust = (32.4/30) * (260*9.81) = 1.08* 2550 = 2754 N - Average acceleration on 5 kg payload: 2754/5 = 550 m/s^2 = 56g — high but brief

Result: Scale 3 achieves orbital insertion with a commercially-sourceable SRM.

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3. Multi-Stage Theoretical Analysis for Large Scale

The Large scale (1000 kg spear) requires three-stage configuration for orbit:

Stage 0: AMSR-UBSCLS → 680 m/s (surface to ~18 km coast) Stage 1: First SRM → additional 3000 m/s (18 km to 200 km coast) Stage 2: Second SRM → remaining 4500 m/s (circularization at 300 km)

Total launch mass: 1000 (spear) + 8000 (Stage 1 SRM) + 2000 (Stage 2 SRM) = 11,000 kg Payload capacity: ~500 kg

This is in the SmallSat/MicroSat launcher class. Comparable to Electron rocket (Virgin Orbit payload: 300 kg to LEO).

Citations (Vol 17)

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- Sutton, G.P., Biblarz, O., Rocket Propulsion Elements, 9th Ed, Wiley 2017
- Turner, M.J.L., Rocket and Spacecraft Propulsion, 3rd Ed, Springer 2009
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